PROJECT REPORT

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Roll No.: 23BEC080

Project title:

a) Design a 4-bit Up/down counter using the Verilog HDL.
b) Generate GDS-II layout of the design using Synopsysflowscripts.
    i. Report Design Area
    ii. Clock frequency, where slack value is positive but less than 1.
    iii. Report power consumption

NOTE: When I designed a 4-bit up/down counter, the leaf cell count was below 20. Hence, I then designed a 16-bit up/down counter and edited the RTL and the testbench code in such a way that the 16-bit counter behaves like a 4-bit counter as the maximum value it reaches is 6, which is within the range of a 4-bit counter i.e., 0 to 15.

1. RTL code

Activities Terminal Jun 3 15:37

student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/rtl

File Edit View Search Terminal Help

```
module updown_counter (
        input Clock,
        input Reset,
        input Enable,
        input UPDN,
        output reg [15:0] COUNT);

reg [15:0] count_next;

always @(posedge Clock) begin
        if (Reset)
                COUNT <= 16'b0;
        else if (Enable)
                COUNT <= count_next;
end

always @(*) begin
        if (UPDN)
                count_next = COUNT + 1;
        else
                count_next = COUNT - 1;
end

endmodule

~
~
~
~
~
~
~
~
~
~
~
"updown_counter_rtl.v" 25L, 460C                                        25,0-1        All
```

2. Testbench code

Activities Terminal Jun 3 15:37

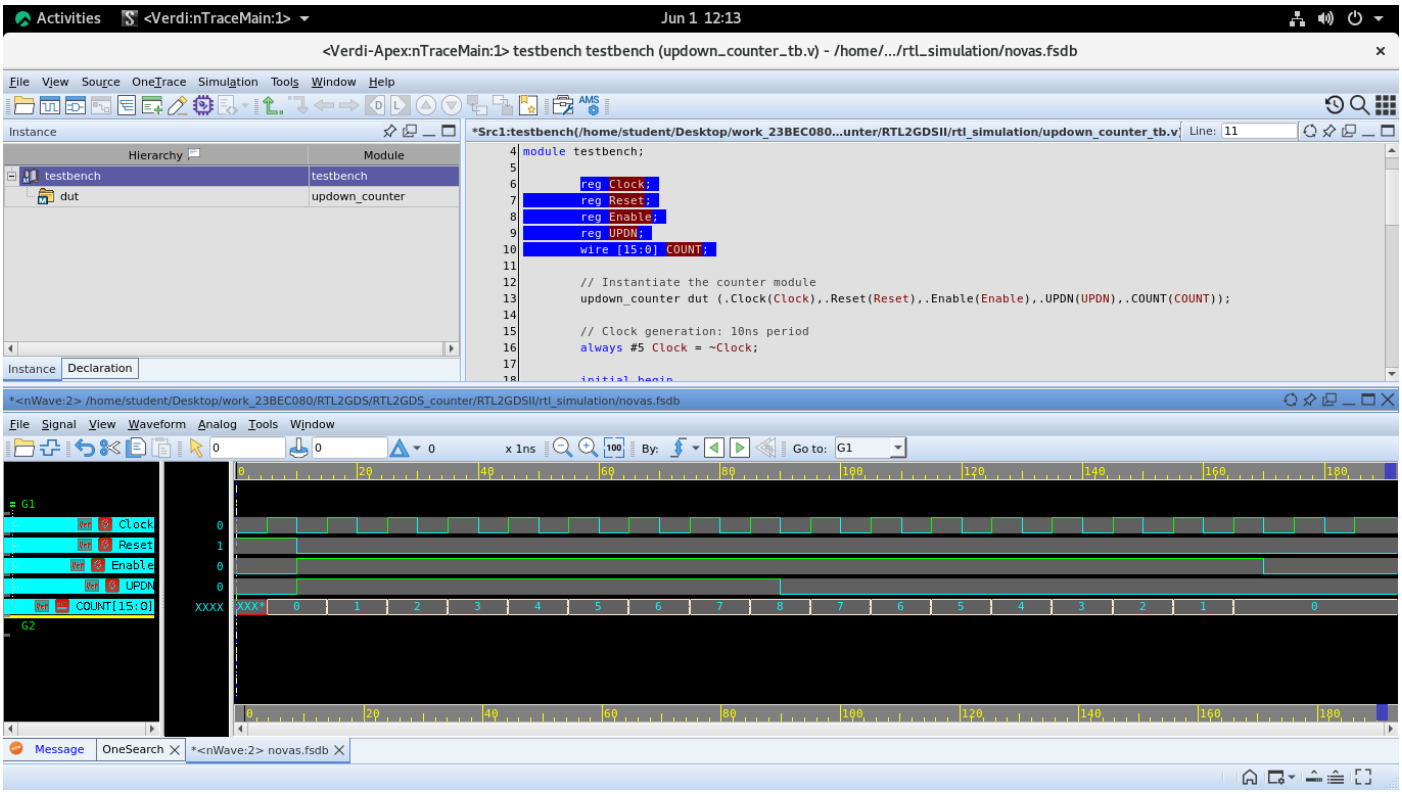
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/rtl_simulation

File Edit View Search Terminal Help

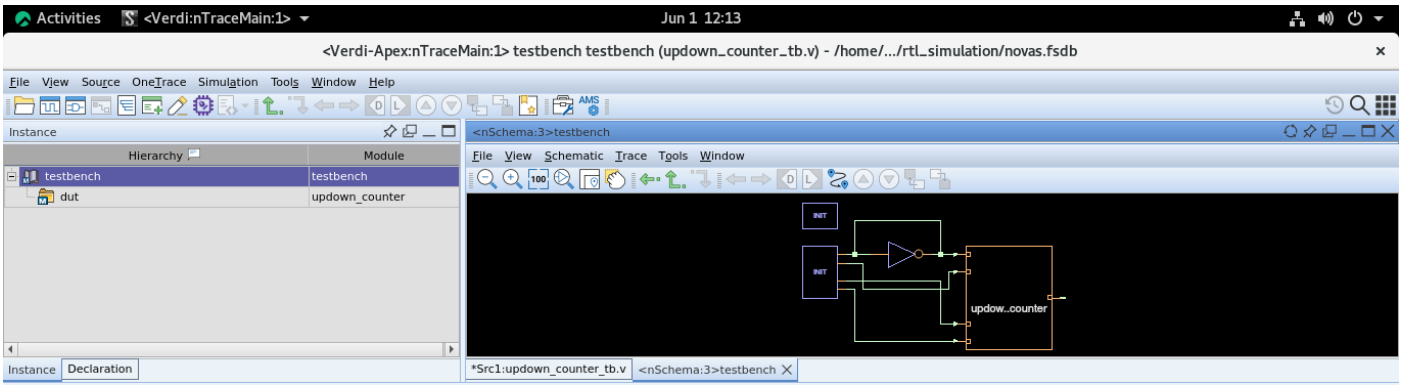
```
`timescale 1ns/1ns
`include "updown_counter_rtl.v" // Includes the RTL code

module testbench;

        reg Clock;
        reg Reset;
        reg Enable;
        reg UPDN;
        wire [15:0] COUNT;

        // Instantiate the counter module
        updown_counter dut (.Clock(Clock),.Reset(Reset),.Enable(Enable),.UPDN(UPDN),.COUNT(COUNT));

        // Clock generation: 10ns period
        always #5 Clock = ~Clock;

        initial begin
                $fsdbDumpvars();

                // Initialize inputs
                Clock = 0;
                Reset = 1;
                Enable = 0;
                UPDN = 0;

                #10 Reset = 0; // Apply reset
                Enable = 1; // Enable counting

                UPDN = 1;
                #80;

                // Change direction to count down
                UPDN = 0;
                #80;


                // Disable counting
                Enable = 0;
                #20;


                $finish;
        end

        // Output monitoring
        initial begin
                $display("Time\tReset\tEnable\tUPDN\tCOUNT");
                $monitor("%0t\t%b\t%b\t%b\t%0d", $time, Reset, Enable, UPDN, COUNT);
        end

endmodule
~
~
                                                                        53,0-1        All
```

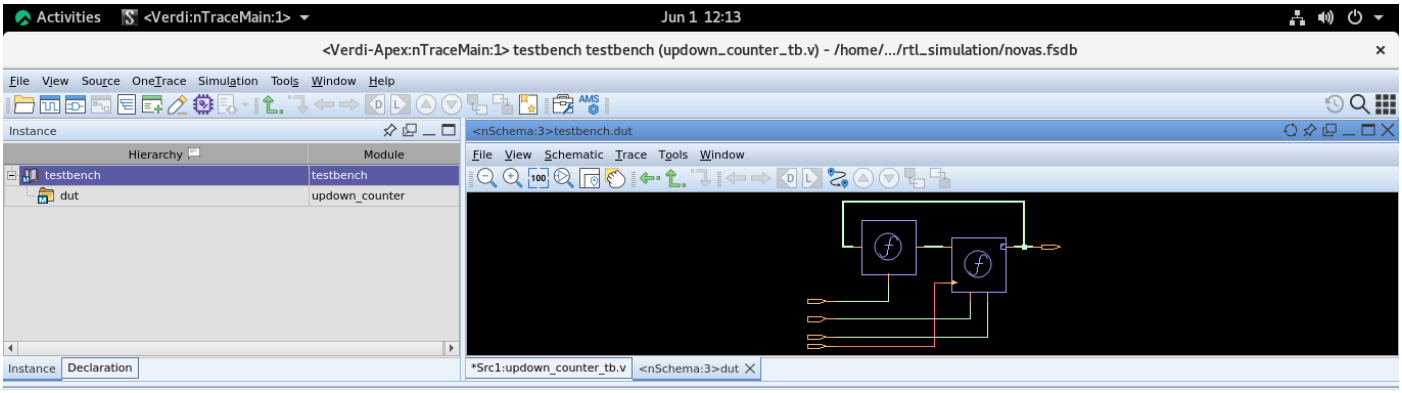
3. Verdi

3.1 Waveform



3.2 Schematics





4. Testbench output (obtained from vcs command)

```
Activities  Terminal                                   Jun 1 12:44
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/DC
File  Edit  View  Search  Terminal  Help
*Verdi* Loading libsscore_vcs202409.so
FSDB Dumper for VCS, Release Verdi_W-2024.09, Linux x86_64/64bit, 08/17/2024
(C) 1996 - 2024 by Synopsys, Inc.
*Verdi* : Create FSDB file 'novas.fsdb'
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : End of traversing.
Time    Reset   Enable  UPDN    COUNT
0       1       0       0       x
5       1       0       0       0
10      0       1       1       0
15      0       1       1       1
25      0       1       1       2
35      0       1       1       3
45      0       1       1       4
55      0       1       1       5
65      0       1       1       6
75      0       1       1       7
85      0       1       1       8
90      0       1       0       8
95      0       1       0       7
105     0       1       0       6
115     0       1       0       5
125     0       1       0       4
135     0       1       0       3
145     0       1       0       2
155     0       1       0       1
165     0       1       0       0
170     0       0       0       0
$finish called from file "updown_counter_tb.v", line 43.
$finish at simulation time                 190
           V C S   S i m u l a t i o n   R e p o r t
Time: 190 ns
CPU Time:      0.180 seconds;       Data structure size:   0.0Mb
Sun Jun  1 12:44:25 2025
[SCL] 06/01/2025 12:44:25 PID:35169 Client:ict-chipin.sot.pdpu.ac.in checkin (null)
[student@ict-chipin rtl_simulation]$ cd ../DC
[student@ict-chipin DC]$
```

5. DESIGN COMPILER

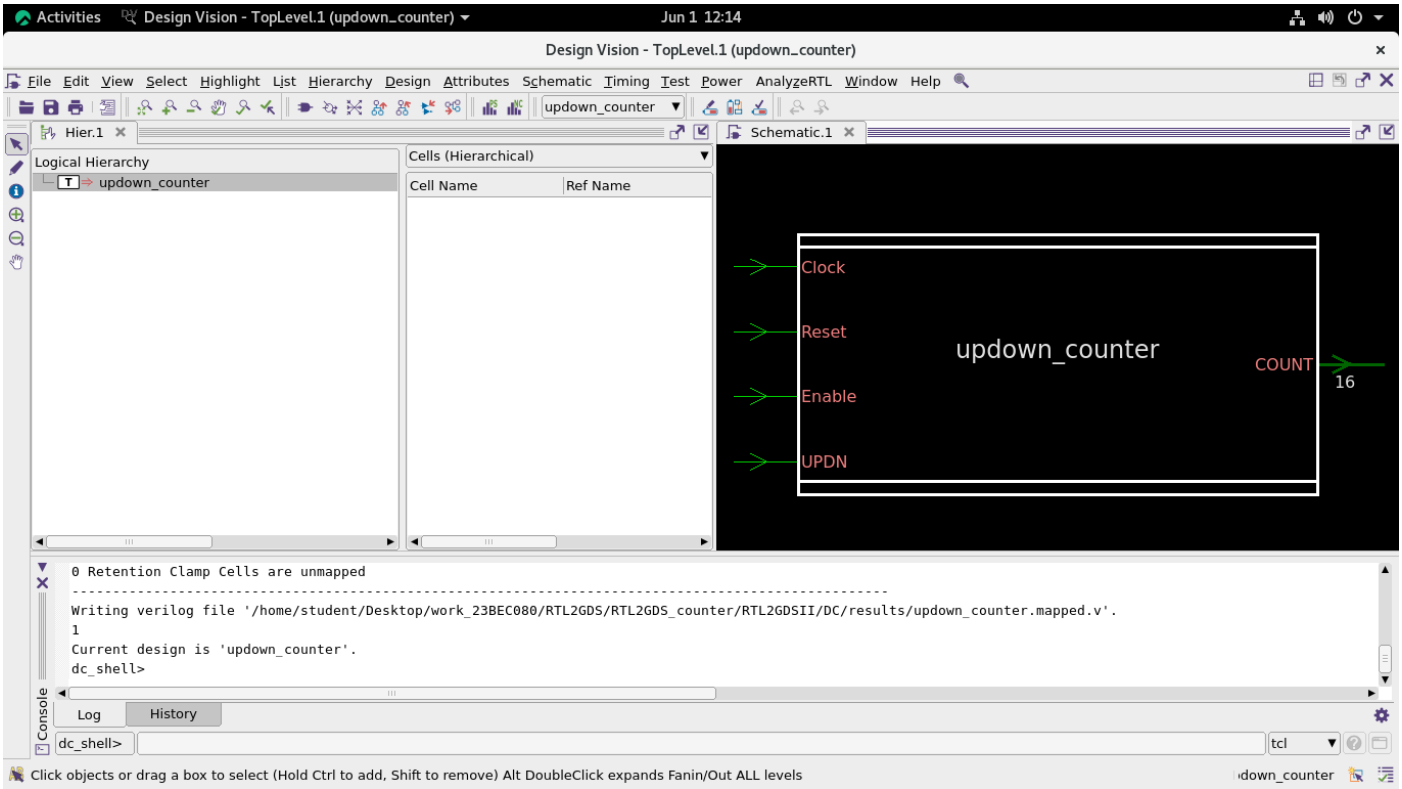
5.1 run_dc.tcl file

```
Activities  Terminal  Jun 3 15:50
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/DC
File  Edit  View  Search  Terminal  Help
source -echo -verbose ./rm_setup/dc_setup.tcl
set RTL_SOURCE_FILES ./../rtl/updown_counter_rtl.v

define_design_lib WORK -path ./WORK

#set_dont_use [get_lib_cells */FADD*]
#set_dont_use [get_lib_cells */HADD*]
#set_dont_use [get_lib_cells */AO*]
#set_dont_use [get_lib_cells */OA*]
#set_dont_use [get_lib_cells */NAND*]
#set_dont_use [get_lib_cells */XOR*]
#set_dont_use [get_lib_cells */NOR*]
#set_dont_use [get_lib_cells */XNOR*]
#set_dont_use [get_lib_cells */MUX*]
#set_dont_use [get_lib_cells */AND*]
#set_dont_use [get_lib_cells */OR*]


analyze -format verilog ${RTL_SOURCE_FILES}
elaborate ${DESIGN_NAME}
current_design


read_sdc ./../CONSTRAINTS/full_adder.sdc

#compile

compile ultra
report_timing
write -format verilog -hierarchy -output ${RESULTS_DIR}/${DCRM_FINAL_VERILOG_OUTPUT_FILE}
~
~
~
~
~
-- INSERT --                                                    31,1          All
```

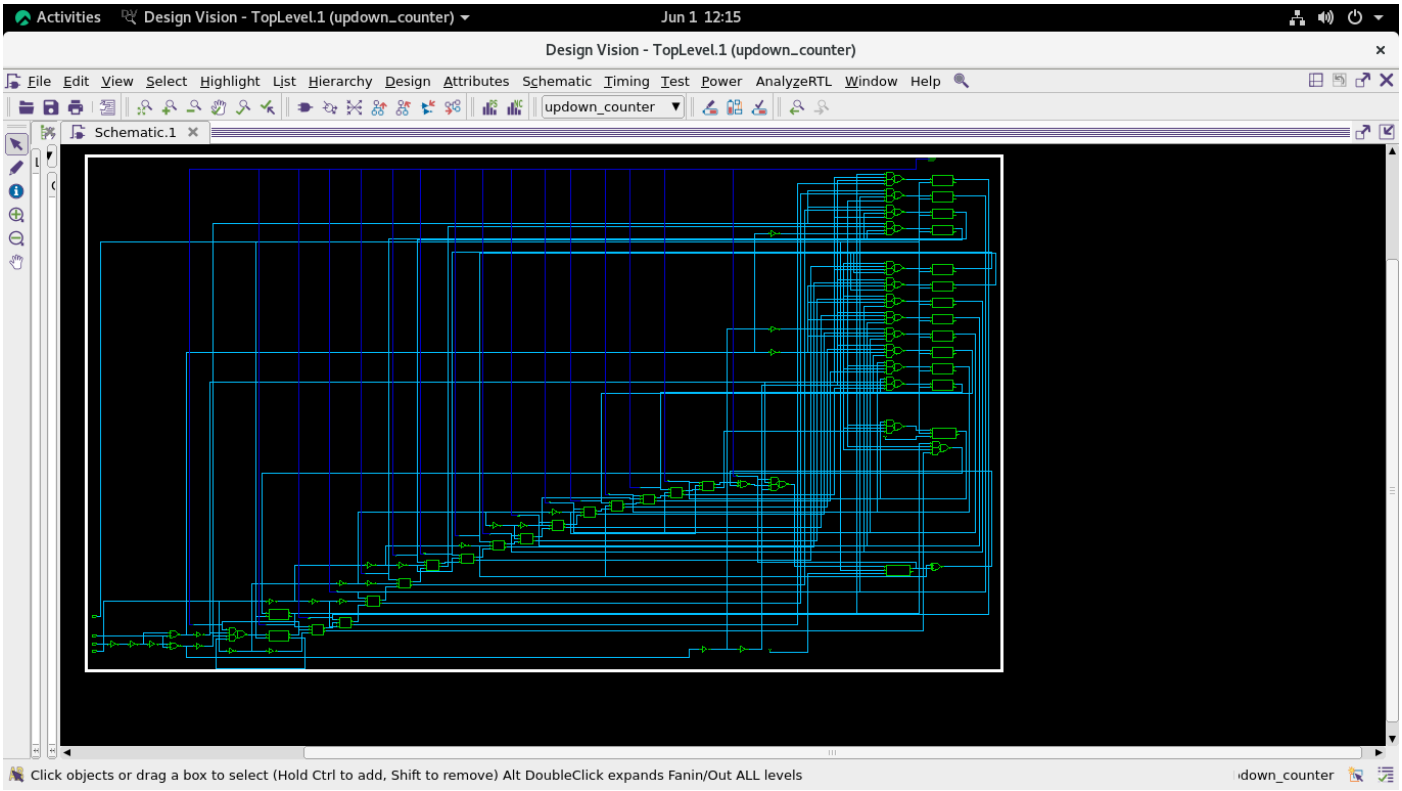
5.2 CONSTRAINTS File - .sdc file

```
Activities  Terminal  Jun 3 15:38
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/CONSTRAINTS
File  Edit  View  Search  Terminal  Help
create_clock -period 2 [get_ports Clock]

set_input_delay -max 0.5 -clock Clock [all_inputs]
set_input_transition 0.5 [all_inputs]

set_output_delay -max 0.5 -clock Clock [all_outputs]

set_clock_uncertainty -setup 0.300 [get_clocks Clock]
set_clock_uncertainty -hold 0.100 [get_clocks Clock]
set_max_transition 0.250 [current_design]
set_max_transition -clock_path 0.150 [get_clocks Clock]

~
~
~
~
~
```

5.3 DC GUI Block diagram



5.4 DC GUI Schematic view



5.5 QOR Report

```
Activities  Terminal                    Jun 1 12:16
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/DC
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...

  Cell Count
  ----------------------------------
  Hierarchical Cell Count:          0
  Hierarchical Port Count:          0
  Leaf Cell Count:                 73
  Buf/Inv Cell Count:              23
  Buf Cell Count:                   6
  Inv Cell Count:                  17
  CT Buf/Inv Cell Count:            0
  Combinational Cell Count:        57
  Sequential Cell Count:           16
  Macro Count:                      0
  ----------------------------------


  Area
  ----------------------------------
  Combinational Area:      155.281982
  Noncombinational Area:   106.740483
  Buf/Inv Area:             33.801153
  Total Buffer Area:            12.20
  Total Inverter Area:          21.60
  Macro/Black Box Area:      0.000000
  Net Area:                 33.963717
  ----------------------------------
  Cell Area:               262.022466
  Design Area:             295.986183


  Design Rules
  ----------------------------------
  Total Number of Nets:            94
  Nets With Violations:             3
```

5.6 Timing Report

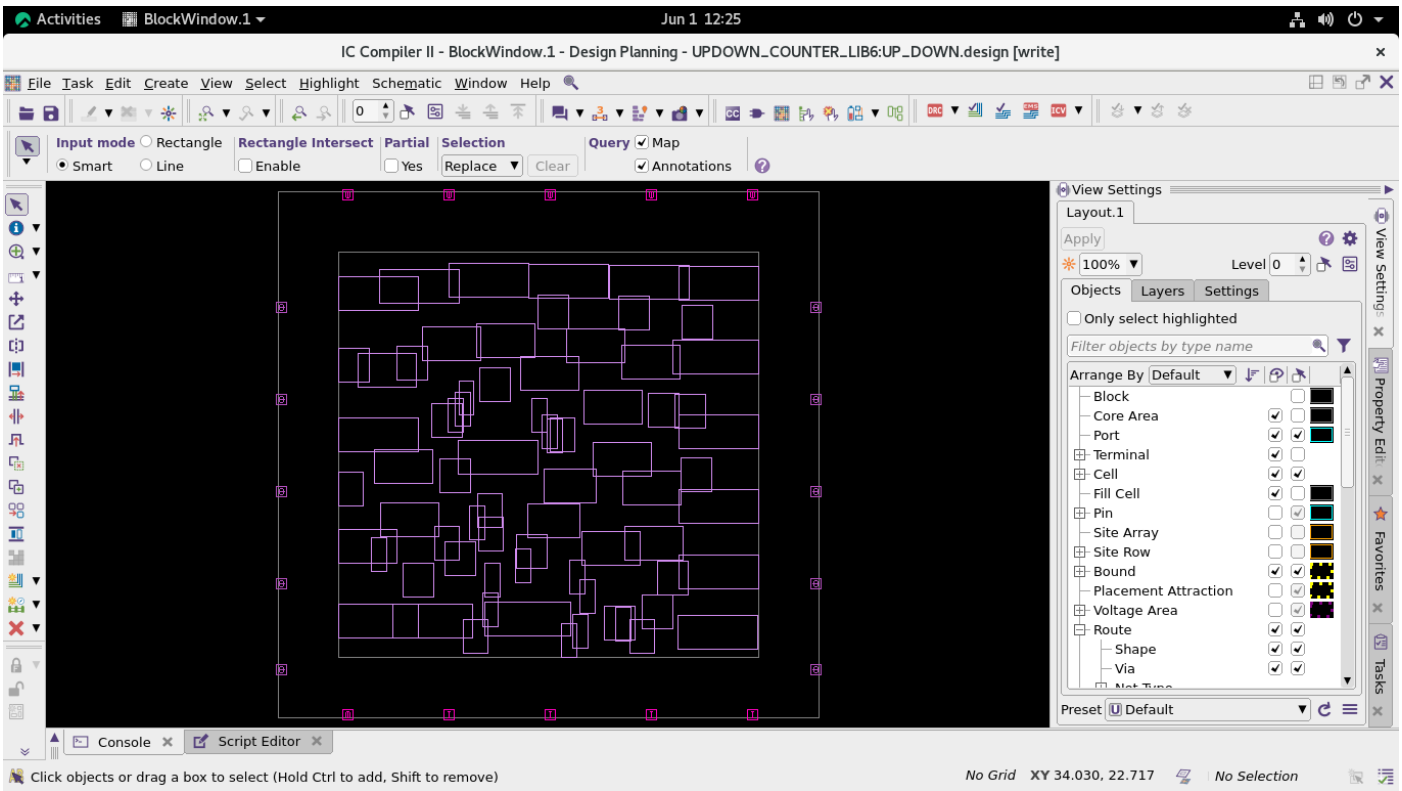
```
Activities  Terminal                    Jun 1 12:22
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/DC
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...

  U33/Y (INVX0_RVT)                        0.07       0.67 r
  U68/CO (FADDX1_RVT)                      0.07       0.74 r
  U60/CO (FADDX1_RVT)                      0.08       0.81 r
  U70/CO (FADDX1_RVT)                      0.07       0.88 r
  U62/CO (FADDX1_RVT)                      0.07       0.95 r
  U71/CO (FADDX1_RVT)                      0.07       1.02 r
  U63/CO (FADDX1_RVT)                      0.07       1.08 r
  U54/CO (FADDX1_RVT)                      0.07       1.15 r
  U65/CO (FADDX1_RVT)                      0.07       1.22 r
  U56/CO (FADDX1_RVT)                      0.07       1.28 r
  U58/CO (FADDX1_RVT)                      0.07       1.35 r
  U72/CO (FADDX1_RVT)                      0.07       1.42 r
  U80/CO (FADDX1_RVT)                      0.07       1.49 r
  U78/CO (FADDX1_RVT)                      0.07       1.55 r
  U76/CO (FADDX1_RVT)                      0.06       1.62 r
  U74/Y (XOR2X1_RVT)                       0.06       1.68 f
  U75/Y (AO22X1_RVT)                       0.03       1.71 f
  COUNT_reg[15]/D (DFFARX1_RVT)            0.01       1.72 f
  data arrival time                                   1.72

  clock Clock (rise edge)                  2.00       2.00
  clock network delay (ideal)              0.00       2.00
  clock uncertainty                       -0.30       1.70
  COUNT_reg[15]/CLK (DFFARX1_RVT)          0.00       1.70 r
  library setup time                      -0.02       1.68
  data required time                                  1.68
  -------------------------------------------------------------
  data required time                                  1.68
  data arrival time                                  -1.72
  -------------------------------------------------------------
  slack (VIOLATED)                                   -0.04


1
dc_shell>
```

6. IC COMPILER II

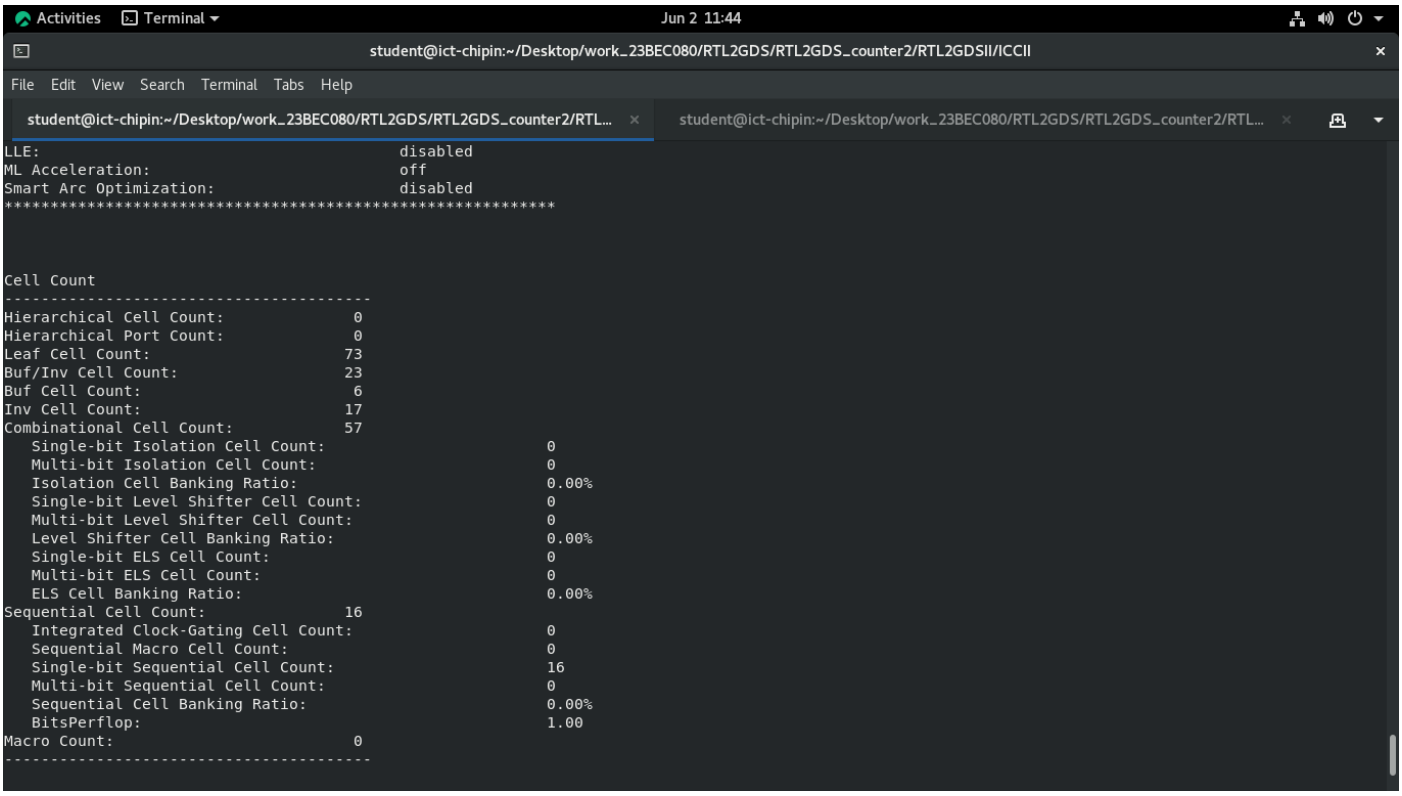
6.1 Floor Planning

Scenario 4:

initialize_floorplan -core_utilization 0.6 -core_offset {3 3} -coincident_boundary false
set_individual_pin_constraints -ports [get_ports B] -sides 2 -pin_spacing_distance 3
place_pins -self
create_placement -floorplan -effort high
save_block -as UP_DOWN
save_lib



6.1.1 Cell count



6.1.2 Area

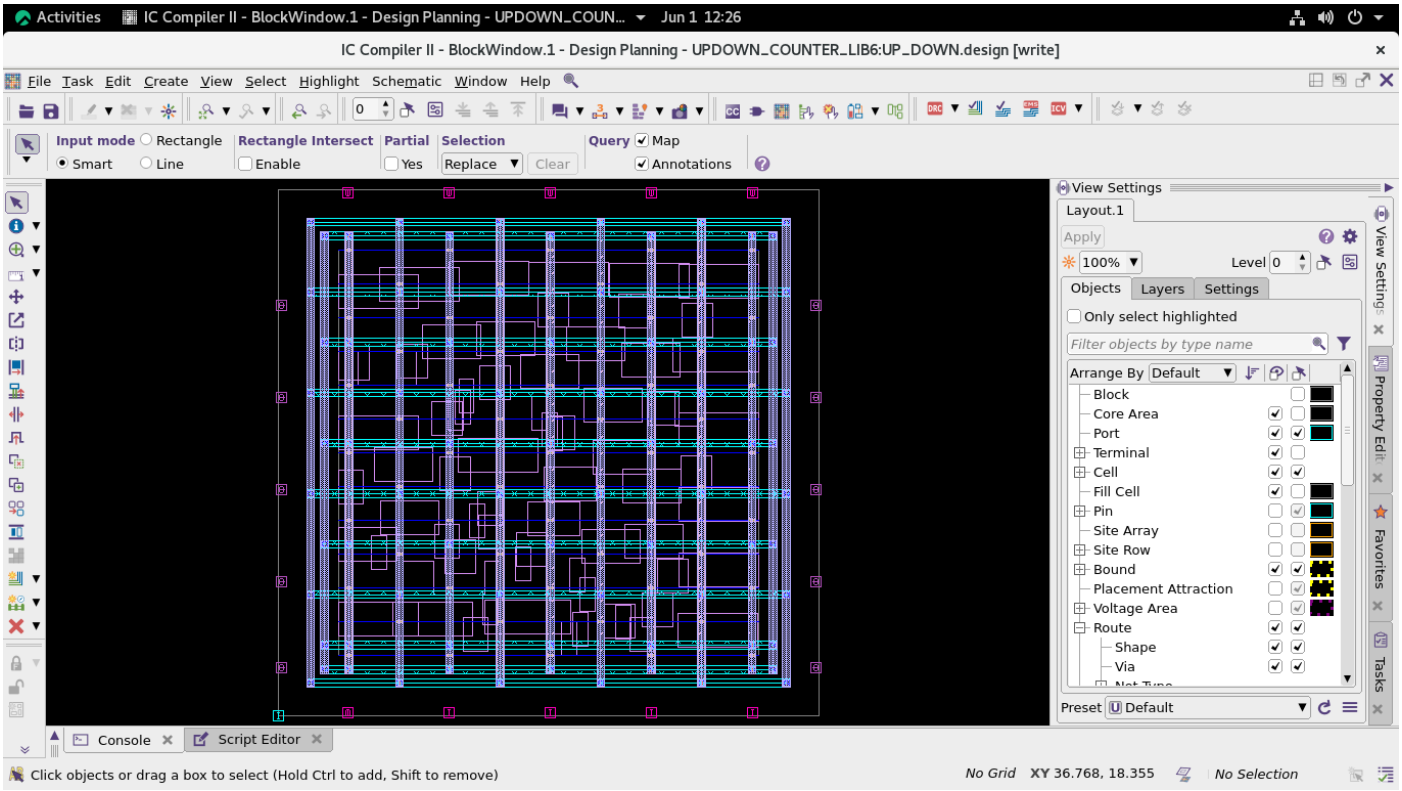
```
Activities  Terminal                    Jun 2 11:44
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...   student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...

   Single-bit Sequential Cell Count:                    16
   Multi-bit Sequential Cell Count:                     0
   Sequential Cell Banking Ratio:                       0.00%
   BitsPerflop:                                         1.00
Macro Count:                          0
---------------------------------------


Area
---------------------------------------
Combinational Area:              155.28
Noncombinational Area:           106.74
Buf/Inv Area:                     33.80
Total Buffer Area:                12.20
Total Inverter Area:              21.60
Macro/Black Box Area:              0.00
Net Area:                             0
Net XLength:                     410.07
Net YLength:                     422.88
---------------------------------------
Cell Area (netlist):                            262.02
Cell Area (netlist and physical only):          262.02
Net Length:                      832.95


Design Rules
---------------------------------------
Total Number of Nets:                93
Nets with Violations:                 0
Max Trans Violations:                 0
Max Cap Violations:                   0
---------------------------------------

1
icc2_shell>
```

6.1.3 Power Report

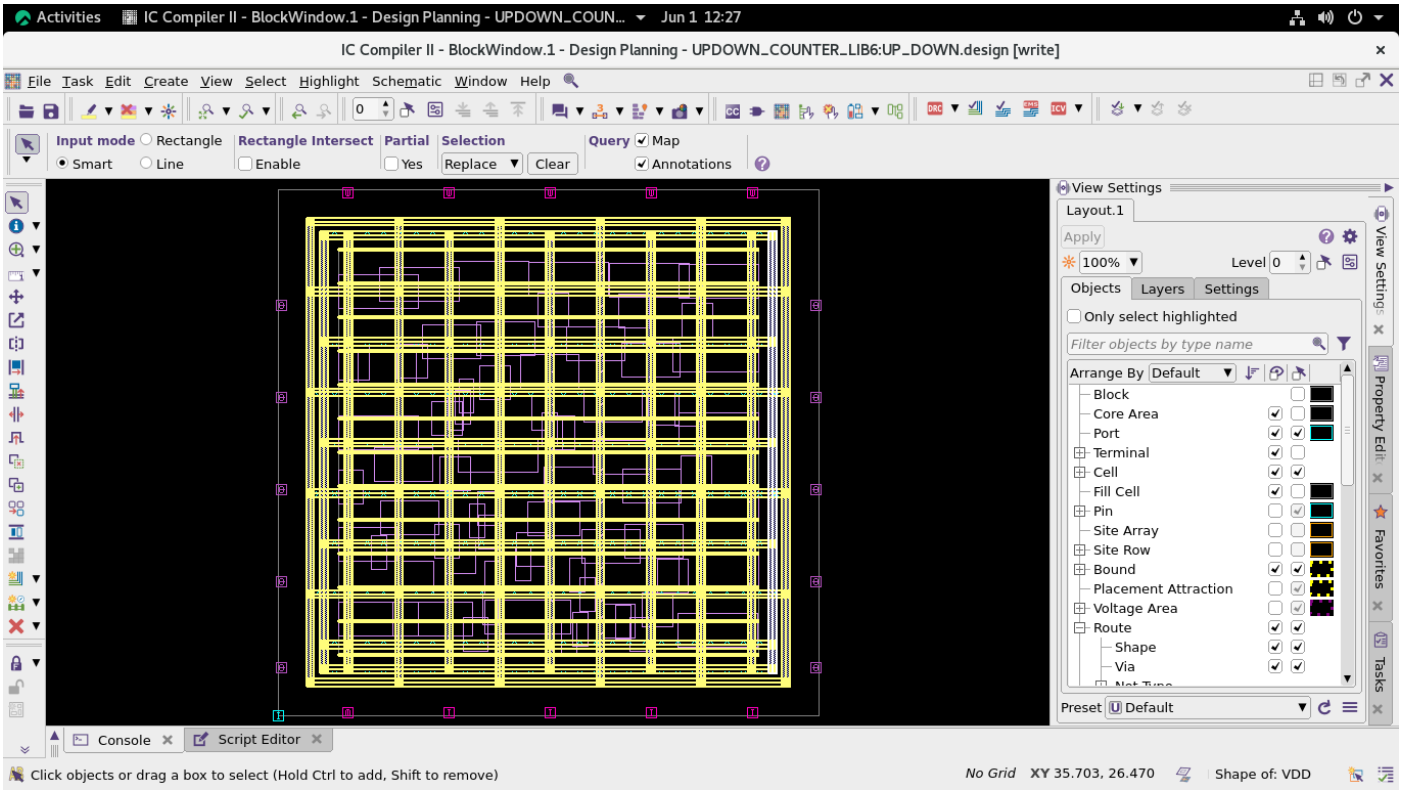
```
Activities  Terminal                    Jun 2 11:44
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...   student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...
alue = inf, value = 0.024090 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port CLK on cell COUNT_reg[12] for parameter Tinp. Lowest table value = 0.016000, highest
table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port Q on cell COUNT_reg[12] for parameter Tinp. Lowest table value = 0.016000, highest ta
ble value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port QN on cell COUNT_reg[12] for parameter Tinp. Lowest table value = 0.016000, highest t
able value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port QN on cell COUNT_reg[12] for parameter Cout. Lowest table value = 0.000100, highest t
able value = 0.008000, value = 0.000000 (POW-046)
Note - message 'POW-046' limit (10) exceeded. Remainder will be suppressed.

  Cell Internal Power    = -1.61e+06 pW (-119.9%)
  Net Switching Power    = 2.96e+06 pW (219.9%)
Total Dynamic Power      = 1.35e+06 pW (100.0%)

Cell Leakage Power       = 5.95e+08 pW


  Attributes
  ----------
      u  -  User defined power group

Power Group         Internal Power        Switching Power          Leakage Power            Total Power    (   %  )    Attrs
--------------------------------------------------------------------------------------------------------------------------------
io_pad                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
memory                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
black_box                0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
clock_network            0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
register                 0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
sequential              -2.79e+05              1.34e+05               3.19e+08               3.19e+08    ( 53.5%)
combinational           -1.33e+06              2.82e+06               2.76e+08               2.77e+08    ( 46.5%)
--------------------------------------------------------------------------------------------------------------------------------
Total                   -1.61e+06 pW           2.96e+06 pW            5.95e+08 pW            5.96e+08 pW
1
icc2_shell>
```

6.1.4 Timing Report: No paths

6.2 Power Planning



Power planning (highlighted)



6.2.1 Cell count

```
Activities  Terminal                                   Jun 1 12:28
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...

Cell Count
---------------------------------------
Hierarchical Cell Count:              0
Hierarchical Port Count:              0
Leaf Cell Count:                     73
Buf/Inv Cell Count:                  23
Buf Cell Count:                       6
Inv Cell Count:                      17
Combinational Cell Count:            57
   Single-bit Isolation Cell Count:                      0
   Multi-bit Isolation Cell Count:                       0
   Isolation Cell Banking Ratio:                         0.00%
   Single-bit Level Shifter Cell Count:                  0
   Multi-bit Level Shifter Cell Count:                   0
   Level Shifter Cell Banking Ratio:                     0.00%
   Single-bit ELS Cell Count:                            0
   Multi-bit ELS Cell Count:                             0
   ELS Cell Banking Ratio:                               0.00%
Sequential Cell Count:               16
   Integrated Clock-Gating Cell Count:                   0
   Sequential Macro Cell Count:                          0
   Single-bit Sequential Cell Count:                     16
   Multi-bit Sequential Cell Count:                      0
   Sequential Cell Banking Ratio:                        0.00%
   BitsPerflop:                                          1.00
Macro Count:                          0
---------------------------------------


Area
---------------------------------------
Combinational Area:              155.28
```

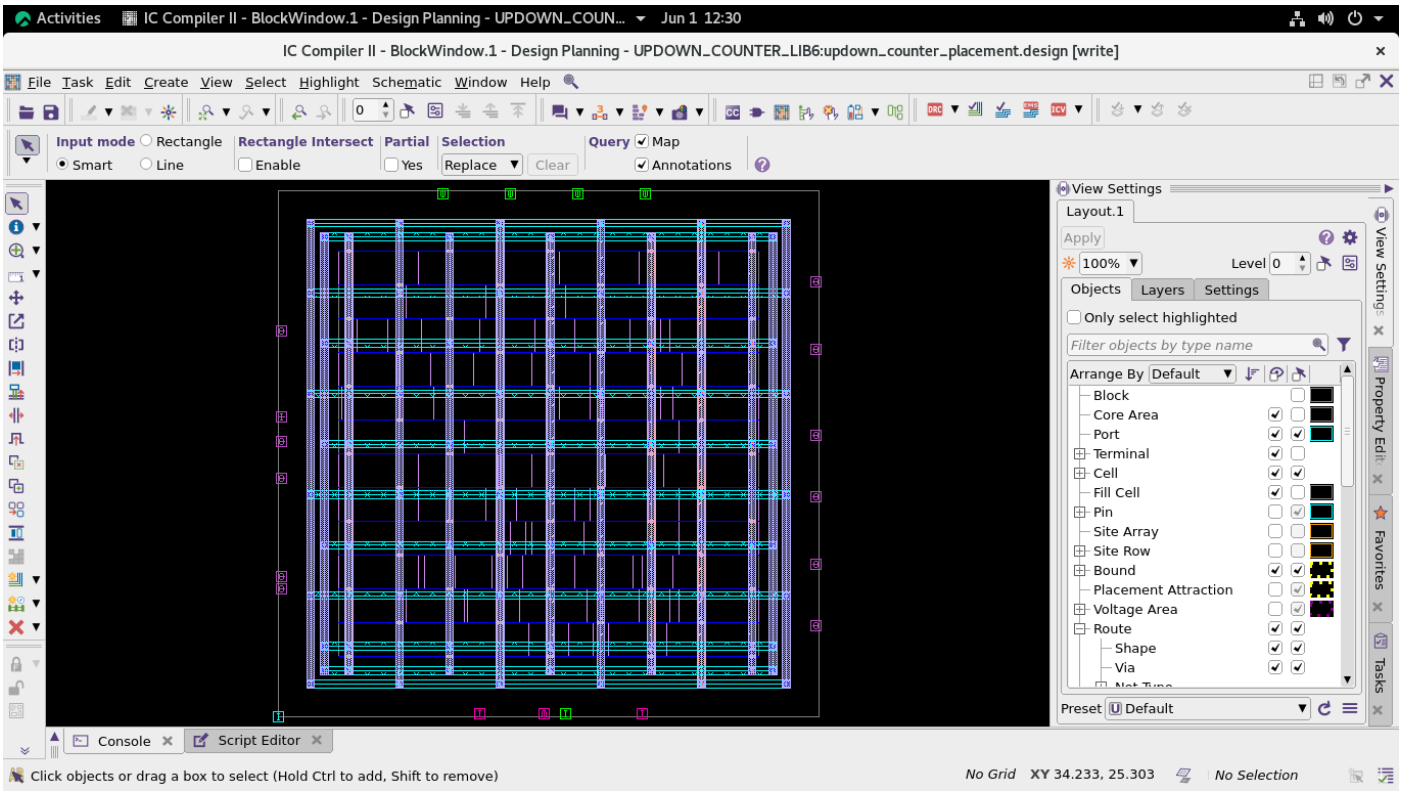
6.2.2 Area

```
Activities  Terminal                                   Jun 1 12:29
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...
   ELS Cell Banking Ratio:                               0.00%
Sequential Cell Count:               16
   Integrated Clock-Gating Cell Count:                   0
   Sequential Macro Cell Count:                          0
   Single-bit Sequential Cell Count:                     16
   Multi-bit Sequential Cell Count:                      0
   Sequential Cell Banking Ratio:                        0.00%
   BitsPerflop:                                          1.00
Macro Count:                          0
---------------------------------------


Area
---------------------------------------
Combinational Area:              155.28
Noncombinational Area:           106.74
Buf/Inv Area:                     33.80
Total Buffer Area:                12.20
Total Inverter Area:              21.60
Macro/Black Box Area:              0.00
Net Area:                             0
Net XLength:                     400.77
Net YLength:                     409.91
---------------------------------------
Cell Area (netlist):                          262.02
Cell Area (netlist and physical only):        262.02
Net Length:                      810.69


Design Rules
---------------------------------------
Total Number of Nets:                94
Nets with Violations:                 0
Max Trans Violations:                 0
Max Cap Violations:                   0
```

6.2.3 Power Report

```
Activities  Terminal                                   Jun 2 11:46
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...   student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...

Temperature Unit     : 1C
Dynamic Power Unit   : 1pW
Leakage Power Unit   : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

  Cell Internal Power    = -1.61e+06 pW (-119.9%)
  Net Switching Power    = 2.96e+06 pW (219.9%)
Total Dynamic Power      = 1.35e+06 pW (100.0%)

Cell Leakage Power       = 5.95e+08 pW


  Attributes
  ----------
      u  -  User defined power group

Power Group         Internal Power        Switching Power         Leakage Power           Total Power    (   %  )    Attrs
--------------------------------------------------------------------------------------------------------------------------------
io_pad                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
memory                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
black_box                0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
clock_network            0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
register                 0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
sequential              -2.79e+05              1.34e+05               3.19e+08               3.19e+08    ( 53.5%)
combinational           -1.33e+06              2.82e+06               2.76e+08               2.77e+08    ( 46.5%)
--------------------------------------------------------------------------------------------------------------------------------
Total                   -1.61e+06 pW           2.96e+06 pW            5.95e+08 pW            5.96e+08 pW
1
icc2_shell>
```

6.3 Placement



6.3.1 Cell count

```
Activities  Terminal                                   Jun 1 12:31
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...
---------------------------------------


Cell Count
---------------------------------------
Hierarchical Cell Count:              0
Hierarchical Port Count:              0
Leaf Cell Count:                     58
Buf/Inv Cell Count:                   8
Buf Cell Count:                       0
Inv Cell Count:                       8
Combinational Cell Count:            42
   Single-bit Isolation Cell Count:                      0
   Multi-bit Isolation Cell Count:                       0
   Isolation Cell Banking Ratio:                         0.00%
   Single-bit Level Shifter Cell Count:                  0
   Multi-bit Level Shifter Cell Count:                   0
   Level Shifter Cell Banking Ratio:                     0.00%
   Single-bit ELS Cell Count:                            0
   Multi-bit ELS Cell Count:                             0
   ELS Cell Banking Ratio:                               0.00%
Sequential Cell Count:               16
   Integrated Clock-Gating Cell Count:                   0
   Sequential Macro Cell Count:                          0
   Single-bit Sequential Cell Count:                     16
   Multi-bit Sequential Cell Count:                      0
   Sequential Cell Banking Ratio:                        0.00%
   BitsPerflop:                                          1.00
Macro Count:                          0
---------------------------------------


Area
---------------------------------------
Combinational Area:              134.19
```

6.3.2 Area

```
Activities  Terminal                                   Jun 1 12:32
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...
   ELS Cell Banking Ratio:                               0.00%
Sequential Cell Count:               16
   Integrated Clock-Gating Cell Count:                   0
   Sequential Macro Cell Count:                          0
   Single-bit Sequential Cell Count:                     16
   Multi-bit Sequential Cell Count:                      0
   Sequential Cell Banking Ratio:                        0.00%
   BitsPerflop:                                          1.00
Macro Count:                          0
---------------------------------------


Area
---------------------------------------
Combinational Area:              134.19
Noncombinational Area:           106.74
Buf/Inv Area:                     13.22
Total Buffer Area:                 0.00
Total Inverter Area:              13.22
Macro/Black Box Area:              0.00
Net Area:                             0
Net XLength:                     322.19
Net YLength:                     358.66
---------------------------------------
Cell Area (netlist):                          240.93
Cell Area (netlist and physical only):        240.93
Net Length:                      680.85


Design Rules
---------------------------------------
Total Number of Nets:                79
Nets with Violations:                 3
Max Trans Violations:                 3
Max Cap Violations:                   0
```

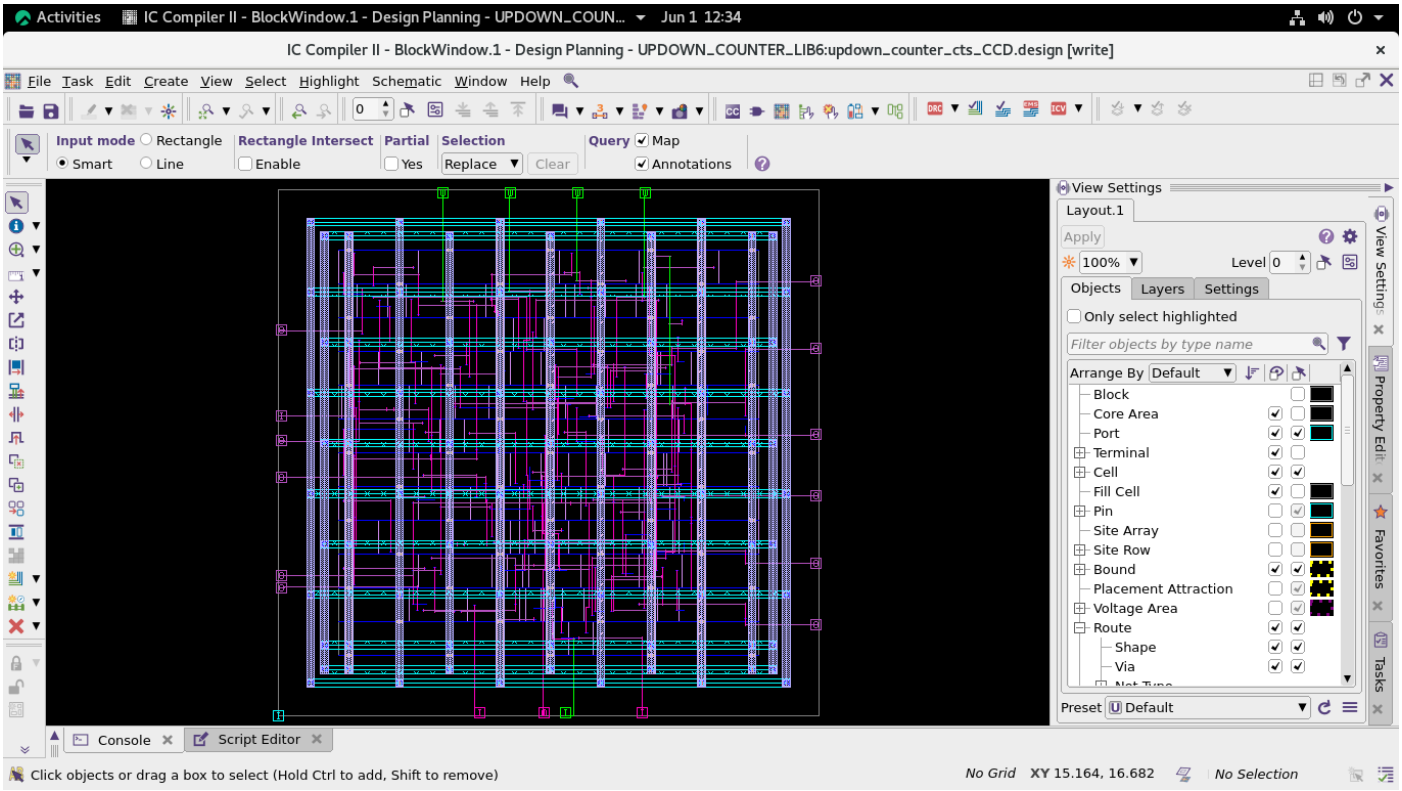
6.3.3 Timing Report

```
Activities  Terminal                                   Jun 1 12:33
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...

  HFSINV_812_1/Y (IBUFFX8_RVT)                    0.12      0.62 r
  U68/CO (FADDX1_RVT)                             0.06      0.68 r
  U60/CO (FADDX1_RVT)                             0.06      0.74 r
  U70/CO (FADDX1_RVT)                             0.05      0.80 r
  U62/CO (FADDX1_RVT)                             0.05      0.85 r
  U71/CO (FADDX1_RVT)                             0.05      0.90 r
  U63/CO (FADDX1_RVT)                             0.05      0.96 r
  U54/CO (FADDX1_RVT)                             0.05      1.01 r
  U65/CO (FADDX1_RVT)                             0.05      1.07 r
  U56/CO (FADDX1_RVT)                             0.06      1.12 r
  U58/CO (FADDX1_RVT)                             0.06      1.18 r
  U72/CO (FADDX1_RVT)                             0.05      1.23 r
  U80/CO (FADDX1_RVT)                             0.05      1.29 r
  U78/CO (FADDX1_RVT)                             0.05      1.34 r
  U76/CO (FADDX1_RVT)                             0.05      1.39 r
  U74/Y (XOR2X1_RVT)                              0.06      1.45 f
  U75/Y (AO22X1_RVT)                              0.03      1.48 f
  COUNT_reg[15]/D (DFFARX1_RVT)                   0.00      1.48 f
  data arrival time                                         1.48

  clock Clock (rise edge)                         2.00      2.00
  clock network delay (ideal)                     0.00      2.00
  COUNT_reg[15]/CLK (DFFARX1_RVT)                 0.00      2.00 r
  clock uncertainty                              -0.30      1.70
  library setup time                             -0.03      1.67
  data required time                                       1.67
  -------------------------------------------------------------------
  data required time                                       1.67
  data arrival time                                       -1.48
  -------------------------------------------------------------------
  slack (MET)                                              0.19


1
icc2_shell>
```

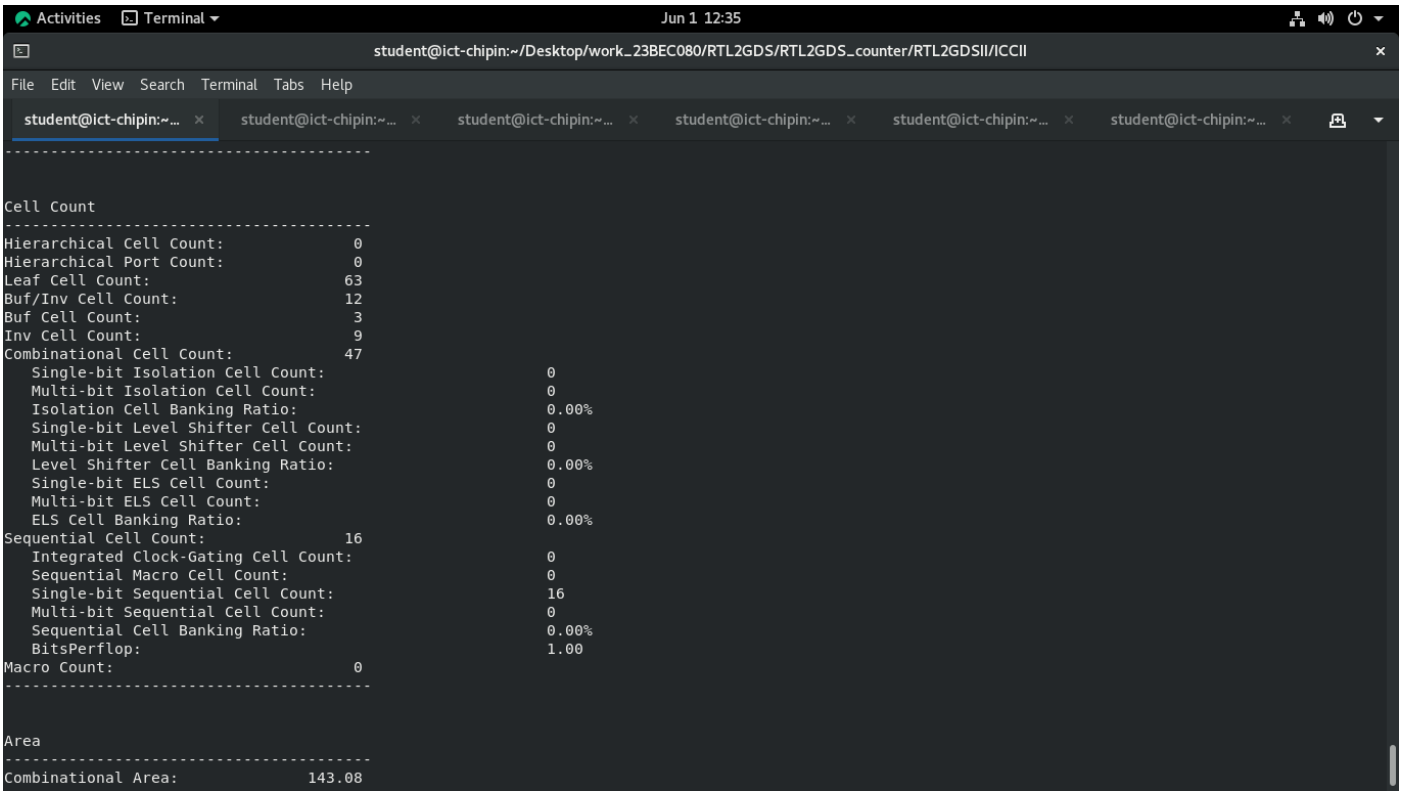
6.3.4 Power Report

```
Activities  Terminal                                   Jun 2 11:49
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...  student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...

Dynamic Power Unit   : 1pW
Leakage Power Unit   : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

  Cell Internal Power    = 2.56e+07 pW ( 89.4%)
  Net Switching Power    = 3.03e+06 pW ( 10.6%)
Total Dynamic Power      = 2.86e+07 pW (100.0%)

Cell Leakage Power       = 5.62e+08 pW


  Attributes
  ----------
      u  -  User defined power group
      i  -  Includes clock pin internal power

Power Group         Internal Power        Switching Power          Leakage Power            Total Power    (   %  )    Attrs
--------------------------------------------------------------------------------------------------------------------------------
io_pad                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
memory                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
black_box                0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
clock_network            3.24e+07              0.00e+00               0.00e+00               3.24e+07    (  5.5%)        i
register                -1.13e+07              7.72e+05               3.20e+08               3.09e+08    ( 52.3%)
sequential               0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
combinational            4.45e+06              2.25e+06               2.42e+08               2.49e+08    ( 42.2%)
--------------------------------------------------------------------------------------------------------------------------------
Total                    2.56e+07 pW           3.03e+06 pW            5.62e+08 pW            5.90e+08 pW
1
icc2_shell>
```

6.4 Clock



6.4.1 Cell count



6.4.2 Area

```
Activities  Terminal                                   Jun 1 12:35
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...

   ELS Cell Banking Ratio:                                0.00%
Sequential Cell Count:                16
   Integrated Clock-Gating Cell Count:                    0
   Sequential Macro Cell Count:                           0
   Single-bit Sequential Cell Count:                      16
   Multi-bit Sequential Cell Count:                       0
   Sequential Cell Banking Ratio:                         0.00%
   BitsPerflop:                                           1.00
Macro Count:                           0
---------------------------------------


Area
---------------------------------------
Combinational Area:              143.08
Noncombinational Area:           106.74
Buf/Inv Area:                     22.87
Total Buffer Area:                 8.39
Total Inverter Area:              14.49
Macro/Black Box Area:              0.00
Net Area:                             0
Net XLength:                     354.21
Net YLength:                     371.90
---------------------------------------
Cell Area (netlist):                           249.82
Cell Area (netlist and physical only):         249.82
Net Length:                      726.11


Design Rules
---------------------------------------
Total Number of Nets:                 84
Nets with Violations:                  4
Max Trans Violations:                  4
Max Cap Violations:                    0
```

6.4.3 Timing Report

```
Activities  Terminal                                   Jun 2 11:51
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...  student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...

  clock network delay (propagated)                0.04      0.04
  input external delay                            0.50      0.54 f
  UPDN (in)                                       0.00      0.54 f
  HFSINV_812_1/Y (IBUFFX8_RVT)                    0.12      0.66 r
  U68/CO (FADDX1_RVT)                             0.06      0.72 r
  U60/CO (FADDX1_RVT)                             0.06      0.78 r
  U70/CO (FADDX1_RVT)                             0.06      0.83 r
  U62/CO (FADDX1_RVT)                             0.06      0.89 r
  U71/CO (FADDX1_RVT)                             0.05      0.94 r
  U63/CO (FADDX1_RVT)                             0.05      1.00 r
  U54/CO (FADDX1_RVT)                             0.05      1.05 r
  U65/CO (FADDX1_RVT)                             0.05      1.10 r
  U56/CO (FADDX1_RVT)                             0.06      1.16 r
  U58/CO (FADDX1_RVT)                             0.05      1.21 r
  U72/CO (FADDX1_RVT)                             0.05      1.27 r
  U80/CO (FADDX1_RVT)                             0.05      1.32 r
  U78/CO (FADDX1_RVT)                             0.05      1.38 r
  U76/CO (FADDX1_RVT)                             0.05      1.43 r
  ctmTdsLR_2_1114/Y (NAND2X0_RVT)                 0.02      1.45 f
  ctmTdsLR_1_1113/Y (OA21X1_RVT)                  0.04      1.49 f
  U75/Y (AO22X1_RVT)                              0.03      1.51 f
  COUNT_reg[15]/D (DFFARX1_RVT)                   0.00      1.51 f
  data arrival time                                         1.51

  clock Clock (rise edge)                         2.00      2.00
  clock network delay (propagated)                0.07      2.07
  COUNT_reg[15]/CLK (DFFARX1_RVT)                 0.00      2.07 r
  clock uncertainty                              -0.30      1.77
  library setup time                             -0.02      1.74
  data required time                                       1.74
  -------------------------------------------------------------------
  data required time                                       1.74
  data arrival time                                       -1.51
  -------------------------------------------------------------------
  slack (MET)                                              0.23
```

6.4.4 Power Report

```
Activities  Terminal                                   Jun 2 11:52
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...   student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...

Dynamic Power Unit   : 1pW
Leakage Power Unit   : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

  Cell Internal Power    = 5.53e+07 pW ( 84.8%)
  Net Switching Power    = 9.92e+06 pW ( 15.2%)
Total Dynamic Power      = 6.52e+07 pW (100.0%)

Cell Leakage Power       = 6.16e+08 pW


  Attributes
  ----------
      u  -  User defined power group
      i  -  Includes clock pin internal power

Power Group         Internal Power        Switching Power          Leakage Power            Total Power    (   %  )    Attrs
--------------------------------------------------------------------------------------------------------------------------------
io_pad                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
memory                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
black_box                0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
clock_network            6.30e+07              6.97e+06               4.93e+07               1.19e+08    ( 17.5%)        i
register                -1.21e+07              7.00e+05               3.20e+08               3.08e+08    ( 45.2%)
sequential               0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
combinational            4.42e+06              2.25e+06               2.48e+08               2.54e+08    ( 37.3%)
--------------------------------------------------------------------------------------------------------------------------------
Total                    5.53e+07 pW           9.92e+06 pW            6.16e+08 pW            6.82e+08 pW
1
icc2_shell>
```

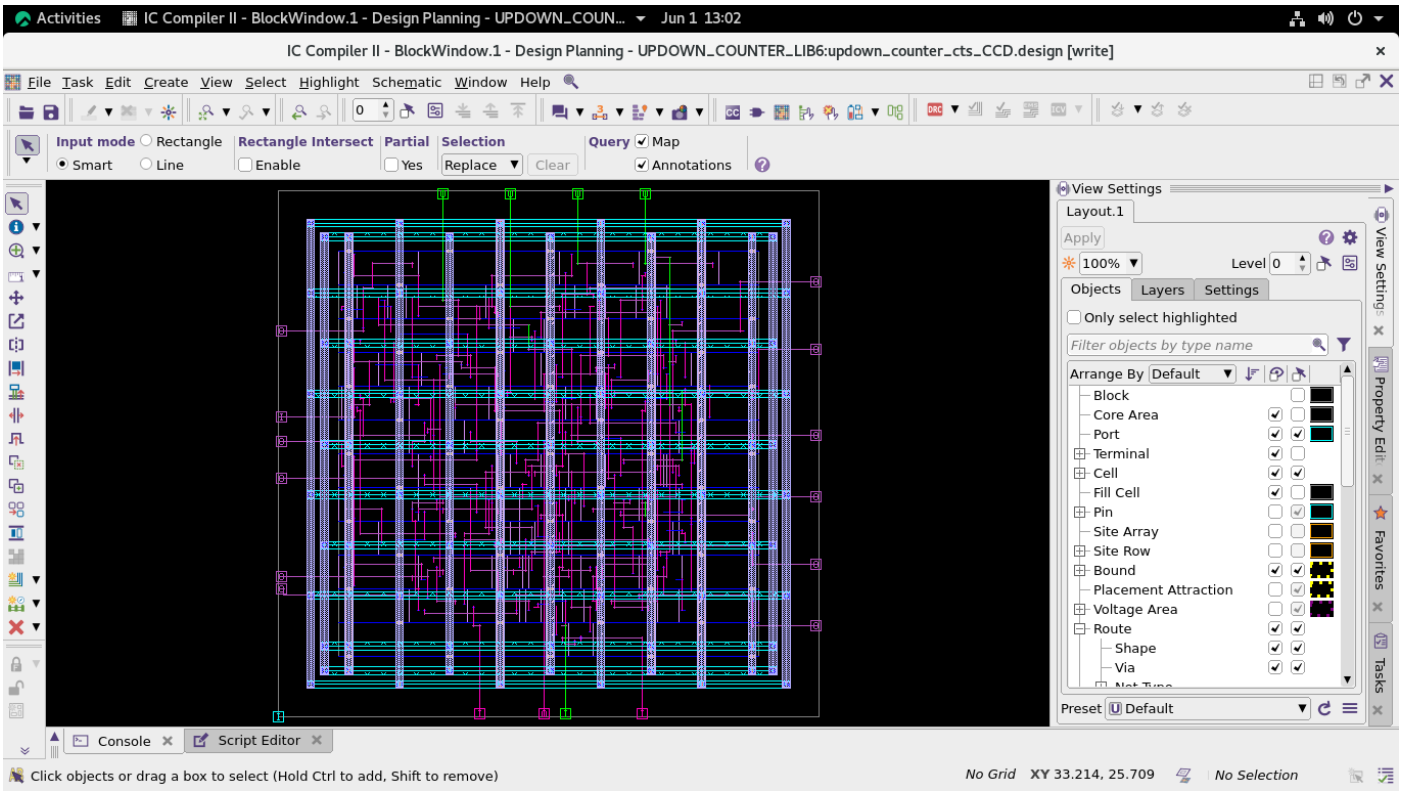
6.4.5 report_clock_qor

```
Activities  Terminal                                   Jun 3 11:05
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...  student@ict-chipin:~...

Design : updown_counter
Version: W-2024.09
Date   : Tue Jun  3 11:04:54 2025
****************************************

Attributes
===========
M Master Clock
G Generated Clock
& Internal Generated Clock
U User Defined Skew Group
D Default Skew Group
* Generated Clock Balanced Separately

========================================
==== Summary Reporting for Corner nom ====
========================================

============================================================ Summary Table for Corner nom ============================================================
==
Clock /                                Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                      Repeater  Repeater  Stdcell   Latency      Skew     Count     Count    Length
                                                                   Count      Area      Area
--------------------------------------------------------------------------------------------------------------------------------------------------
--
### Mode: func, Scenario: func::nom
Clock                                   M,D        16      3        2      6.35      6.35      0.07      0.00         1         0     79.28
--------------------------------------------------------------------------------------------------------------------------------------------------
--
All Clocks                                         16      3        2      6.35      6.35      0.07      0.00         1         0     79.28


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
1
icc2_shell>
```

6.4.6 report_clock_qor -largest 2 -show_verbose_paths > cts_ccd.rpt

```
Largest Path #2
Mode                : func
Corner              : nom
Scenario            : func::nom
Skew Group          : default_Clock
Clock Fanout        : Clock
Clock at Sink       : Clock
Sink                : COUNT_reg[11]/CLK
Latency             : 0.07
-----------------------------------------------

  Point                                       Fanout   Cap    Trans    Incr     Path          Location          NDR                    Layers
        CTS Tags
  ---------------------------------------------------------------------------------------------------------------------------------------------------
---------------------
  source latency                                                          0.00     0.00
  Clock (in)                                     1      1.27    0.50     0.00     0.00 r    (0.14,14.86)      default                        0
        clk,trans
  ctosc_drc_inst_1110/A (NBUFFX4_RVT)                                     0.50     0.00     0.00 r    (3.80,17.31)                                 0
            trans
  ctosc_drc_inst_1110/Y (NBUFFX4_RVT)            1      1.88    0.11     0.03     0.03 r    (4.46,17.21)      default                        0

  ctosc_drc_inst_995/A (NBUFFX8_RVT)                                      0.11     0.00     0.03 r    (6.91,15.44)                                 0

  ctosc_drc_inst_995/Y (NBUFFX8_RVT)            16     17.41    0.04     0.04     0.07 r    (7.96,15.54)      default            {M2 31} {M3 30}

  COUNT_reg[11]/CLK (DFFX1_RVT)                                           0.04     0.00     0.07 r    (20.51,9.52)                       {M2 31} {M3 30}
            sink
  ---------------------------------------------------------------------------------------------------------------------------------------------------
---------------------
  total clock latency                                                                      0.07


1
icc2_shell>
```

6.5 Routing



6.5.1 Cell count

```
Activities  Terminal                                   Jun 1 13:03
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Help
Total Hold Violation:              0.00
No. of Hold Violations:               0
----------------------------------------


Cell Count
----------------------------------------
Hierarchical Cell Count:              0
Hierarchical Port Count:              0
Leaf Cell Count:                     63
Buf/Inv Cell Count:                  12
Buf Cell Count:                       3
Inv Cell Count:                       9
Combinational Cell Count:            47
   Single-bit Isolation Cell Count:                      0
   Multi-bit Isolation Cell Count:                       0
   Isolation Cell Banking Ratio:                         0.00%
   Single-bit Level Shifter Cell Count:                  0
   Multi-bit Level Shifter Cell Count:                   0
   Level Shifter Cell Banking Ratio:                     0.00%
   Single-bit ELS Cell Count:                            0
   Multi-bit ELS Cell Count:                             0
   ELS Cell Banking Ratio:                               0.00%
Sequential Cell Count:               16
   Integrated Clock-Gating Cell Count:                   0
   Sequential Macro Cell Count:                          0
   Single-bit Sequential Cell Count:                     16
   Multi-bit Sequential Cell Count:                      0
   Sequential Cell Banking Ratio:                        0.00%
   BitsPerflop:                                          1.00
Macro Count:                          0
----------------------------------------


Area
----------------------------------------
Combinational Area:             143.08
```

6.5.2 Area

```
Activities  Terminal                                   Jun 1 13:03
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Help
   Single-bit ELS Cell Count:                            0
   Multi-bit ELS Cell Count:                             0
   ELS Cell Banking Ratio:                               0.00%
Sequential Cell Count:               16
   Integrated Clock-Gating Cell Count:                   0
   Sequential Macro Cell Count:                          0
   Single-bit Sequential Cell Count:                     16
   Multi-bit Sequential Cell Count:                      0
   Sequential Cell Banking Ratio:                        0.00%
   BitsPerflop:                                          1.00
Macro Count:                          0
----------------------------------------


Area
----------------------------------------
Combinational Area:             143.08
Noncombinational Area:          106.74
Buf/Inv Area:                    22.87
Total Buffer Area:                8.39
Total Inverter Area:             14.49
Macro/Black Box Area:             0.00
Net Area:                            0
Net XLength:                    352.80
Net YLength:                    383.93
----------------------------------------
Cell Area (netlist):                          249.82
Cell Area (netlist and physical only):        249.82
Net Length:                     736.73


Design Rules
----------------------------------------
Total Number of Nets:                84
Nets with Violations:                 4
Max Trans Violations:                 4
Max Cap Violations:                   0
```

6.5.3 Timing Report

```
Activities  Terminal                                   Jun 1 13:03
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Help
  UPDN (in)                                          0.00      0.54 f
  HFSINV_812_1/Y (IBUFFX8_RVT)                       0.12      0.66 r
  U68/CO (FADDX1_RVT)                                0.06      0.72 r
  U60/CO (FADDX1_RVT)                                0.06      0.78 r
  U70/CO (FADDX1_RVT)                                0.05      0.83 r
  U62/CO (FADDX1_RVT)                                0.05      0.89 r
  U71/CO (FADDX1_RVT)                                0.05      0.94 r
  U63/CO (FADDX1_RVT)                                0.05      0.99 r
  U54/CO (FADDX1_RVT)                                0.05      1.05 r
  U65/CO (FADDX1_RVT)                                0.05      1.10 r
  U56/CO (FADDX1_RVT)                                0.06      1.16 r
  U58/CO (FADDX1_RVT)                                0.05      1.21 r
  U72/CO (FADDX1_RVT)                                0.05      1.27 r
  U80/CO (FADDX1_RVT)                                0.05      1.32 r
  U78/CO (FADDX1_RVT)                                0.05      1.38 r
  U76/CO (FADDX1_RVT)                                0.05      1.43 r
  ctmTdsLR_2_1114/Y (NAND2X0_RVT)                    0.02      1.45 f
  ctmTdsLR_1_1113/Y (OA21X1_RVT)                     0.04      1.48 f
  U75/Y (AO22X1_RVT)                                 0.03      1.51 f
  COUNT_reg[15]/D (DFFARX1_RVT)                      0.00      1.51 f
  data arrival time                                            1.51

  clock Clock (rise edge)                            2.00      2.00
  clock network delay (propagated)                   0.07      2.07
  COUNT_reg[15]/CLK (DFFARX1_RVT)                    0.00      2.07 r
  clock uncertainty                                 -0.30      1.77
  library setup time                                -0.02      1.74
  data required time                                           1.74
  -------------------------------------------------------------------------
  data required time                                           1.74
  data arrival time                                           -1.51
  -------------------------------------------------------------------------
  slack (MET)                                                  0.23


1
icc2_shell>
```

6.5.4 Power Report

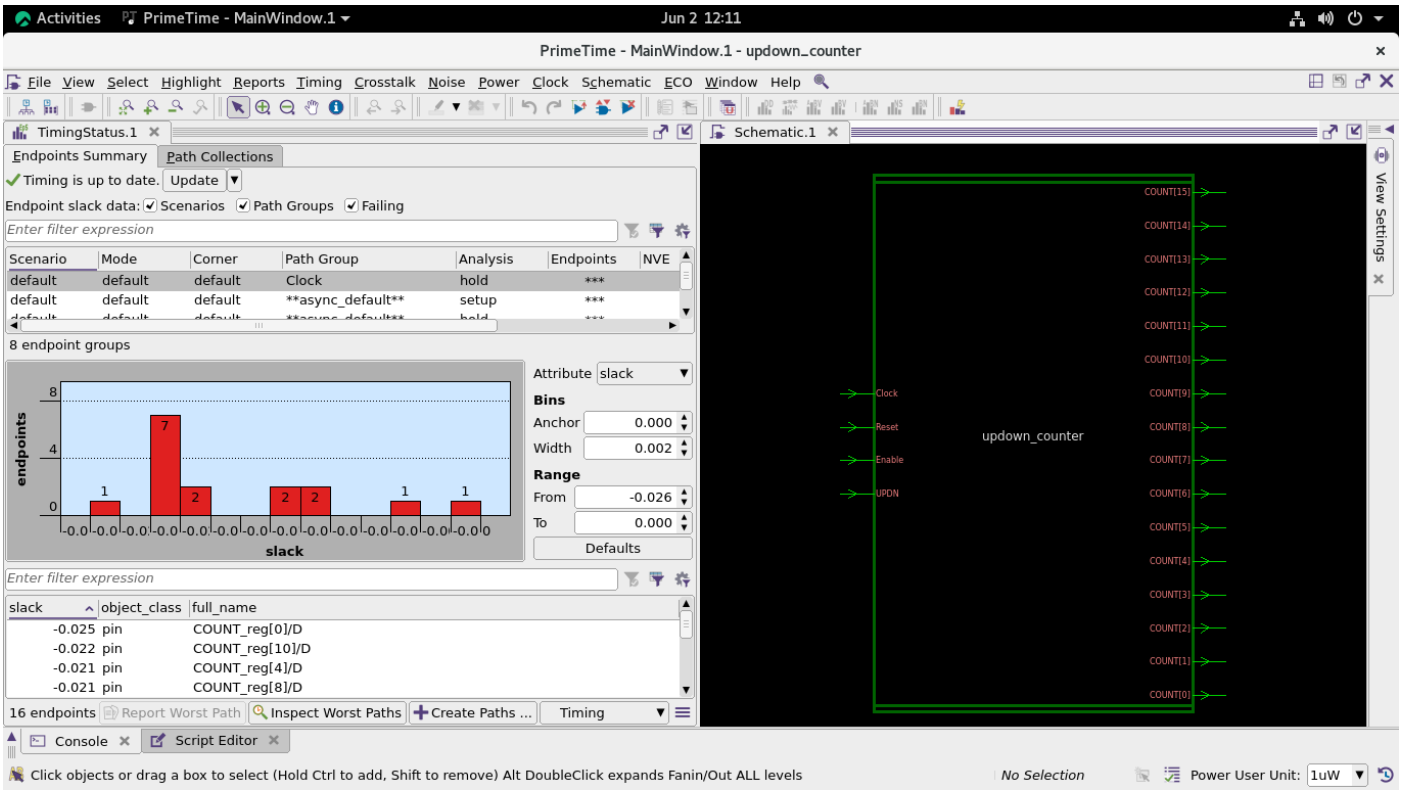
```
Activities  Terminal                                   Jun 2 11:52
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/ICCII
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...   student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...
Dynamic Power Unit   : 1pW
Leakage Power Unit   : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

  Cell Internal Power    = 5.53e+07 pW ( 84.9%)
  Net Switching Power    = 9.83e+06 pW ( 15.1%)
Total Dynamic Power      = 6.51e+07 pW (100.0%)

Cell Leakage Power       = 6.16e+08 pW


  Attributes
  ----------
      u  -  User defined power group
      i  -  Includes clock pin internal power

Power Group         Internal Power        Switching Power          Leakage Power            Total Power    (   %  )    Attrs
--------------------------------------------------------------------------------------------------------------------------------
io_pad                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
memory                   0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
black_box                0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
clock_network            6.30e+07              6.88e+06               4.93e+07               1.19e+08    ( 17.5%)        i
register                -1.20e+07              7.04e+05               3.20e+08               3.08e+08    ( 45.3%)
sequential               0.00e+00              0.00e+00               0.00e+00               0.00e+00    (  0.0%)
combinational            4.37e+06              2.25e+06               2.47e+08               2.54e+08    ( 37.2%)
--------------------------------------------------------------------------------------------------------------------------------
Total                    5.53e+07 pW           9.83e+06 pW            6.16e+08 pW            6.81e+08 pW
1
icc2_shell>
```

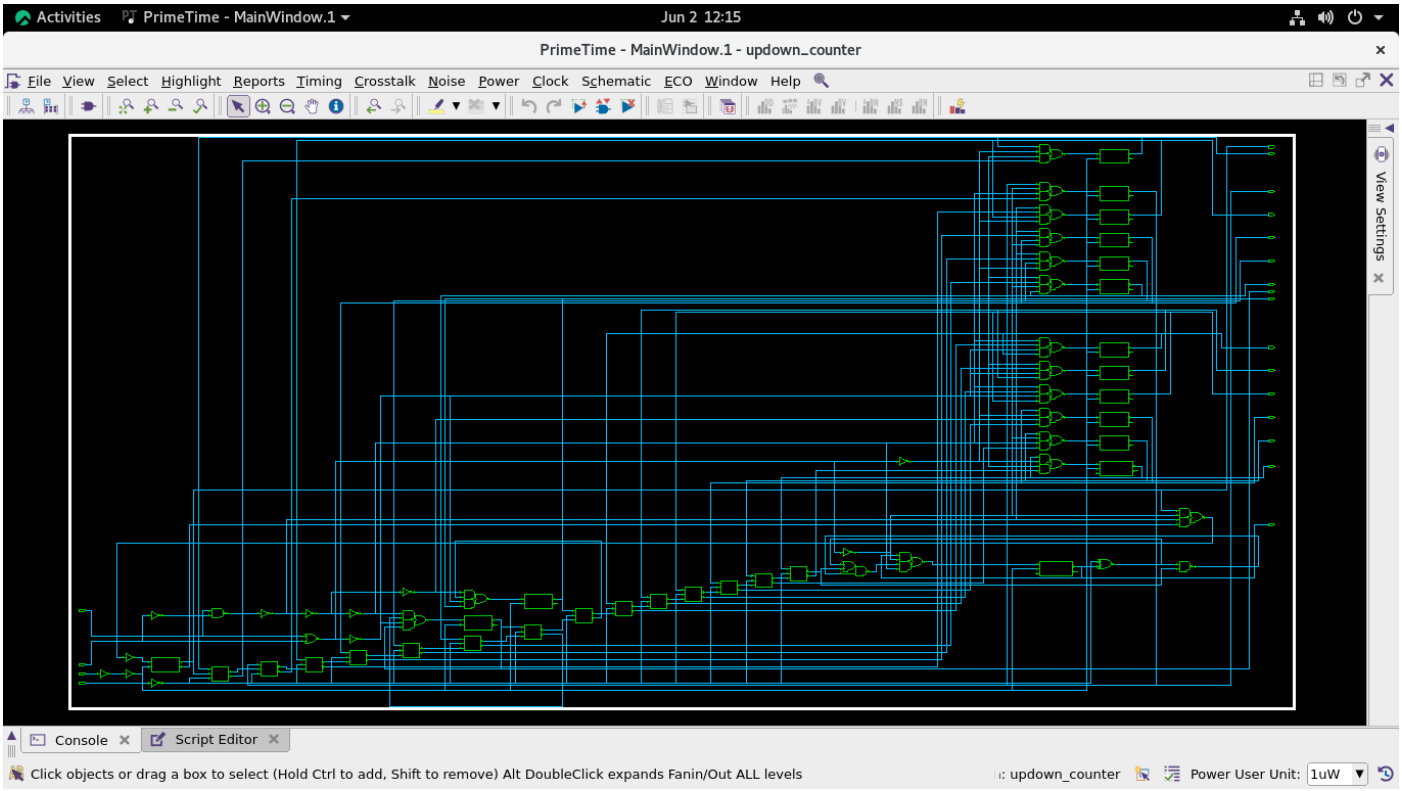
7. PRIMETIME

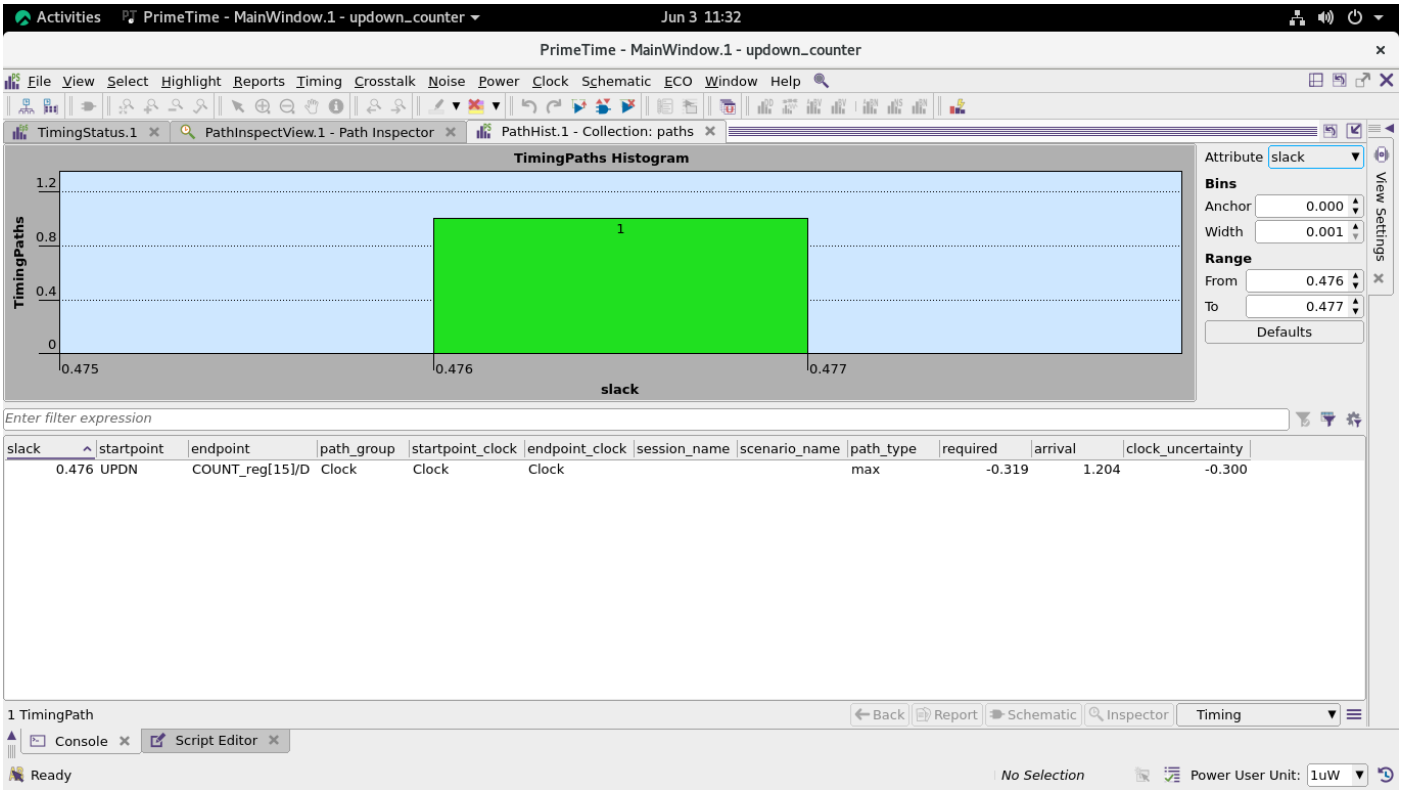
7.1 Final timing report

```
Activities  Terminal                                   Jun 1 13:10
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL2GDSII/PT
File  Edit  View  Search  Terminal  Tabs  Help
student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...   student@ict-chipin:~/Desktop/work_23BEC080/RTL2GDS/RTL2GDS_counter2/RTL...

  U60/CO (FADDX1_RVT)                                0.041193 & 0.660990 r
  U70/CO (FADDX1_RVT)                                0.041031 & 0.702022 r
  U62/CO (FADDX1_RVT)                                0.040748 & 0.742769 r
  U71/CO (FADDX1_RVT)                                0.039686 & 0.782455 r
  U63/CO (FADDX1_RVT)                                0.039799 & 0.822254 r
  U54/CO (FADDX1_RVT)                                0.039344 & 0.861598 r
  U65/CO (FADDX1_RVT)                                0.040366 & 0.901964 r
  U56/CO (FADDX1_RVT)                                0.042491 & 0.944455 r
  U58/CO (FADDX1_RVT)                                0.040242 & 0.984697 r
  U72/CO (FADDX1_RVT)                                0.040310 & 1.025007 r
  U80/CO (FADDX1_RVT)                                0.040559 & 1.065565 r
  U78/CO (FADDX1_RVT)                                0.039947 & 1.105512 r
  U76/CO (FADDX1_RVT)                                0.040921 & 1.146433 r
  ctmTdsLR_2_1114/Y (NAND2X0_RVT)                    0.013600 & 1.160034 f
  ctmTdsLR_1_1113/Y (OA21X1_RVT)                     0.025204 & 1.185238 f
  U75/Y (AO22X1_RVT)                                 0.019145 & 1.204383 f
  COUNT_reg[15]/D (DFFARX1_RVT)                      0.000005 & 1.204387 f
  data arrival time                                             1.204387

  clock Clock (rise edge)                            2.000000   2.000000
  clock network delay (ideal)                        0.000000   2.000000
  COUNT_reg[15]/CLK (DFFARX1_RVT)                               2.000000 r
  clock reconvergence pessimism                      0.000000   2.000000
  clock uncertainty                                 -0.300000   1.700000
  library setup time                                -0.019383   1.680617
  data required time                                            1.680617
  -------------------------------------------------------------------------
  data required time                                            1.680617
  data arrival time                                            -1.204387
  -------------------------------------------------------------------------
  slack (MET)                                                   0.476230


1
pt_shell>
```

7.2 GUI schematics and graph:




Activities
PrimeTime - MainWindow.1
Jun 2 12:15
PrimeTime - MainWindow.1 - updown_counter
File View Select Highlight Reports Timing Crosstalk Noise Power Clock Schematic ECO Window Help
View Settings
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove) Alt DoubleClick expands Fanin/Out ALL levels
: updown_counter
Power User Unit: 1uW


Activities
PrimeTime - MainWindow.1 - updown_counter
Jun 3 11:32
PrimeTime - MainWindow.1 - updown_counter
File View Select Highlight Reports Timing Crosstalk Noise Power Clock Schematic ECO Window Help
TimingStatus.1
PathInspectView.1 - Path Inspector
PathHist.1 - Collection: paths
TimingPaths Histogram
TimingPaths
1.2
0.8
0.4
0
1
0.475
0.476
0.477
slack
Attribute slack
Bins
Anchor 0.000
Width 0.001
Range
From 0.476
To 0.477
Defaults
View Settings
Enter filter expression
slack startpoint endpoint path_group startpoint_clock endpoint_clock session_name scenario_name path_type required arrival clock_uncertainty
0.476 UPDN COUNT_reg[15]/D Clock Clock Clock max -0.319 1.204 -0.300
1 TimingPath
Back Report Schematic Inspector Timing
Console
Script Editor
Ready
No Selection
Power User Unit: 1uW

LEARNINGS AND OUTCOMES:

While working on this project, I was able to gain hands-on experience in designing a counter under certain given constraints such as physical constraints and timing constraints. I learned how to verify the functionality of my code using Verdi. I was also able to perform synthesis analysis using the Synopsys Design Compiler tool and I also explored the entire physical design in IC Compiler II inclusive of floor planning, power planning, placement, clock and routing. Along with this, I analysed how the design area, leaf cell count of the design, slack as well as power consumption changes at every stage of the design. This project enhanced my understanding of digital design, verification, and industry-standard EDA tools used in VLSI workflows.

Changes made:

1. CONSTRAINTS: In *full_adder.sdc* file, the clock period was changed from 1 to 2.
2. DC:
    a. *rm_setup*: In *common_setup.tcl*, the library file (.db file) was changed from tt to ff in TARGET LIBRARY FILES section.
    b. In the *run_dc.tcl* file, the name of the RTL source file was changed from *full_adder_rtl.v* to *updown_counter_rtl.v*
    c. The name of the netlist file was automatically changed from *full_adder.mapped.v* to *updown_counter.mapped.v* after sourcing the *run_dc.tcl* file in the design compiler.
3. ICCII: In *floorplan.tcl*, the name of the mapped file was changed from *full_adder.mapped.v* to *updown_counter.mapped.v*. The design name is changed from *full_adder* to *updown_counter*. The name of the library and block is also changed accordingly.
4. PT: In *run_pt_p1.tcl* file, the name of the standard cell library is changed from tt to ff.